Attendance Code: 381967



UNIVERSITY OF LINCOLN

CMP2028M

Natural Language Processing

Week 4: “Vector Semantics and Embeddings”

Heriberto Cuayahuitl

https://staff.Lincoln.ac.uk/hcuayahuitl

Agenda

- Week 1, 5-6 Feb: Tokenization and regular expressions
- Week 2, 12-13 Feb: N-gram language models
- Week 3, 19-20 Feb: Text classification
- Week 4, 27-27 Feb: **Vector semantics and embeddings**
- Week 5, 5-6 Mar: Large Language Models (LLMs)
- Week 6, 12-13 Mar: Question answering & information retrieval
- Week 7, 19-20 Mar: Constituent and dependency parsing
- Week 8, 26-27 Mar: Information extraction
- Week 9, 23-24 Apr: Overview of neural networks
- Week 10, 30 Apr-1May: Machine translation
- Week 11, 7-8 May: Chatbots and dialogue systems
- Week 12, 14-15 May: Speech recognition and synthesis

Easter break

Last Week

- **Supervised classification** is useful for a wide number of text-based applications with features $\mathbf{x} = \{x_1, x_2, \ldots, x_n\}$ and labels y. The later being binary $(+, -)$ or multiclass $(-$, neutral, $+$, etc.)

- There are many **methods** available, probabilistic and neural-based are only some of them.
  - N-gram classifiers
  - Logistic regression
  - Multi-layer perceptrons

Today

Vector semantics

Word and Sentence Embeddings

Text similarity

Embedding-Based Supervised Classification

Acknowledgement: Some images from https://web.stanford.edu/~jurafsky/slp3/5.pdf

Vector Semantics

"Vector semantics" is the theory for representing meaning as vectors of numbers—the standard way to represent word meaning in NLP.

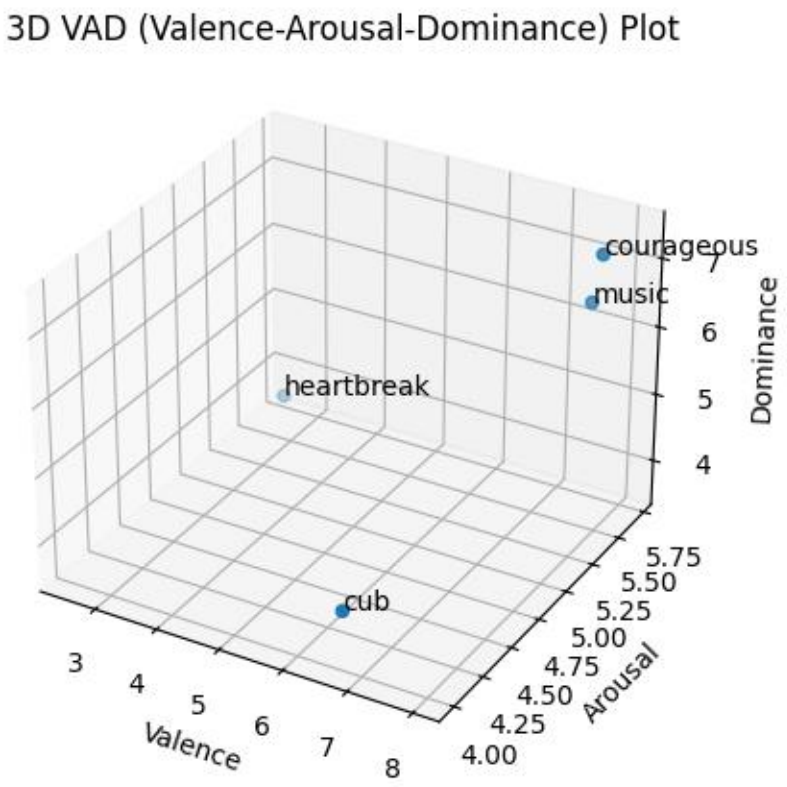
Osgood et al., 1957 found that words varied along three important dimensions of affective meaning:

	Valence	Arousal	Dominance
courageous	8.0	5.5	7.4
music	7.7	5.6	6.5
heartbreak	2.5	5.7	3.6
cub	6.7	4.0	4.2

- Words like happy or satisfied are high on **valence** (the opposite for unhappy or annoyed).
- Excited is high on arousal, while calm is low on **arousal**.
- Controlling is high on dominance, while influenced is low on dominance

Vector Semantics (cont.)

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The idea that word meaning can be represented as a point in a high-dimensional space (beyond 3D) is the foundation of vector semantics.

Vector Semantics: Example

Suppose we don't know the meaning of the word ongchoi but we see it in the following contexts:

- Ongchoi is delicious sauteed with garlic.
- Ongchoi is superb over rice.
- ...ongchoi leaves with salty sauces...

Suppose also that we know the following:

- ...spinach sauteed with garlic over rice...
- ...chard stems and leaves are delicious...
- ...collard greens and other salty leafy greens

The above suggests that ongchoi is a leafy green.

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Count-Based Word Embeddings

This is based on a **co-occurrence matrix** of dimensionality $|V| * |V|$.

Each cell records the number of times the row and column words (target and context, respectively) co-occur nearby in a training corpus.

A context window around the word, for example 4 words to the left and 4 words to the right.

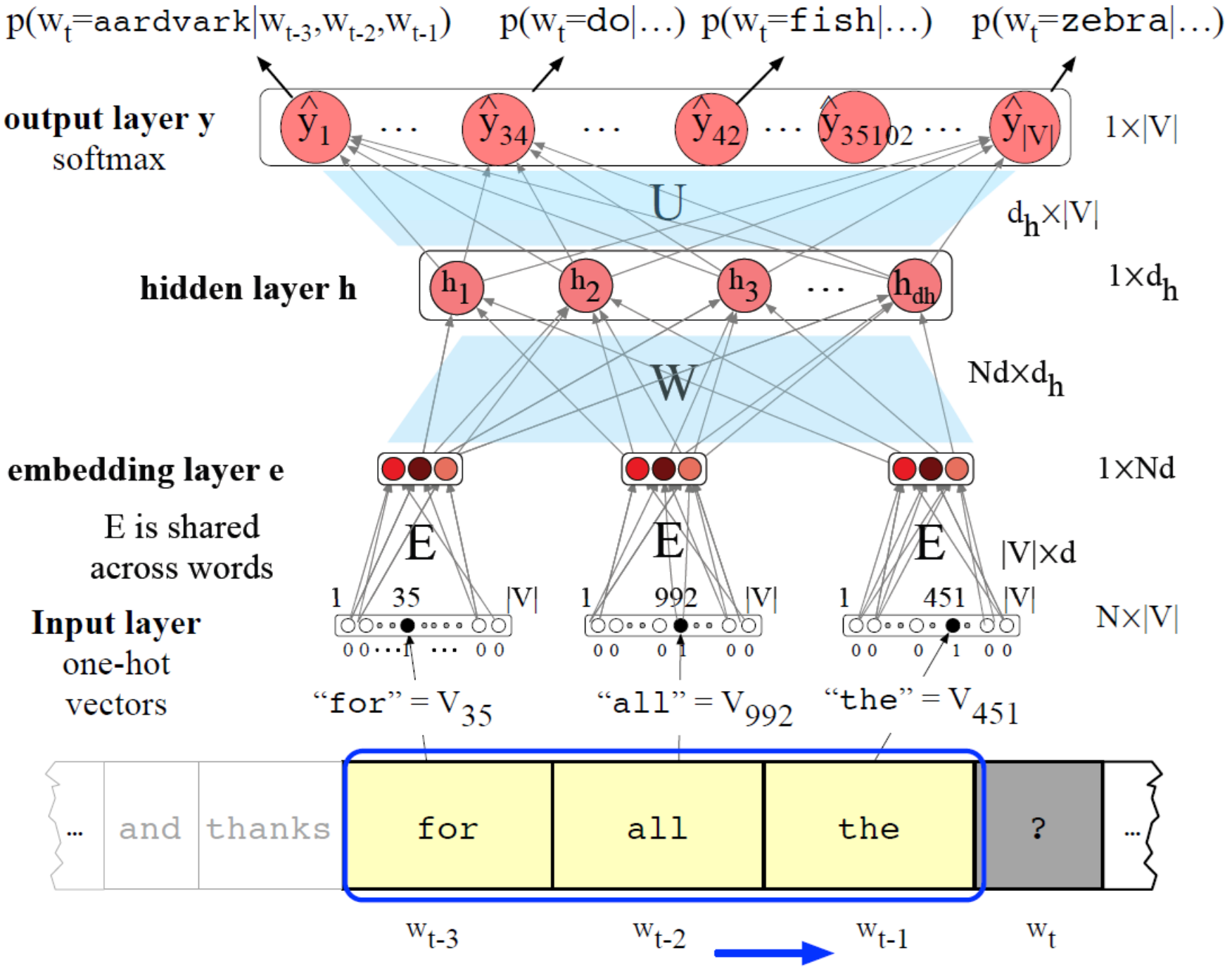
Example

is traditionally followed by	**cherry**	pie, a traditional dessert
often mixed, such as	**strawberry**	rhubarb pie. Apple pie
computer peripherals and personal	**digital**	assistants. These devices usually
a computer. This includes	**information**	available on the internet

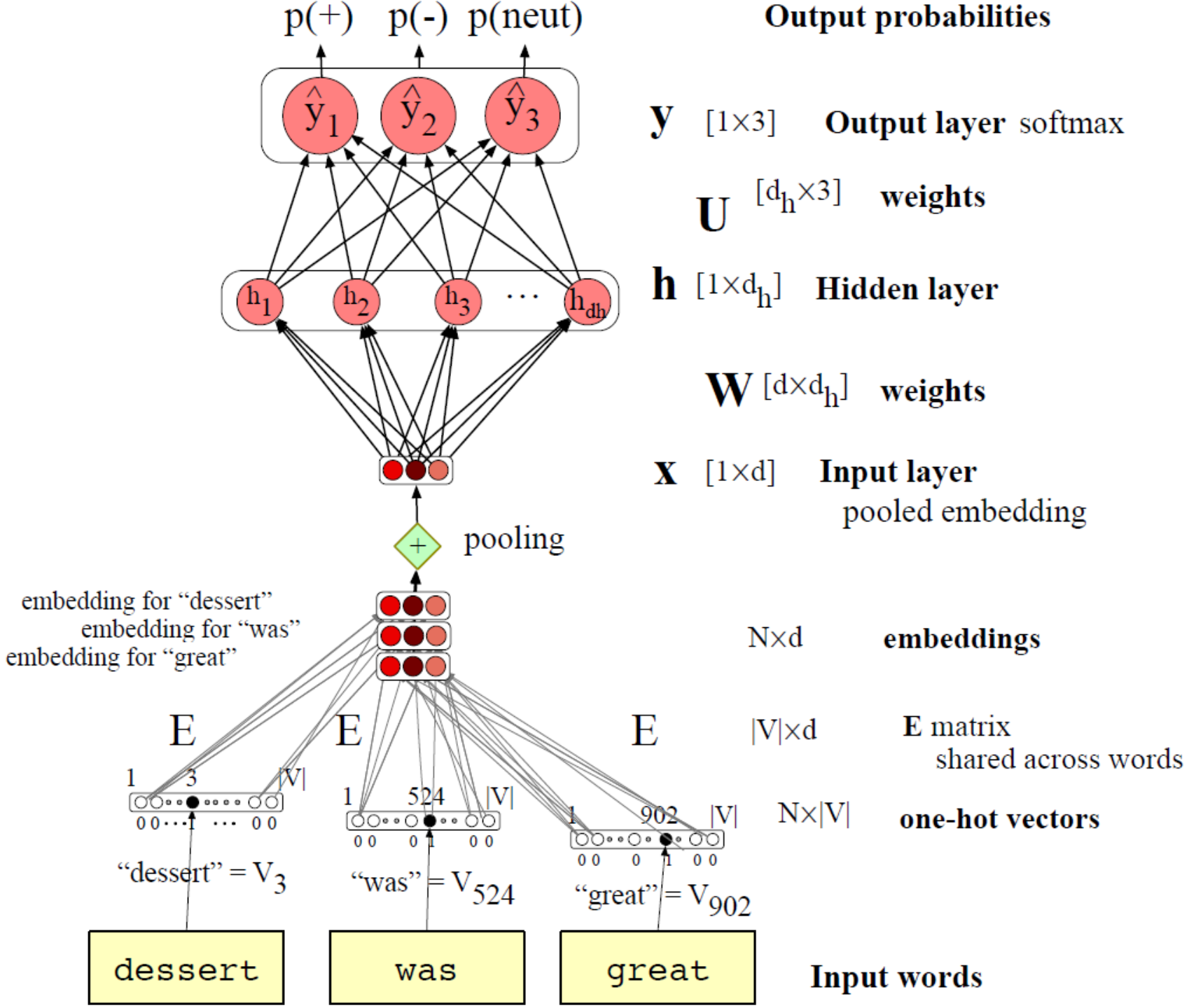
	a	**computer**	**pie**
cherry	1	0	1
strawberry	0	0	2
digital	0	1	0
information	1	1	0

Co-occurrence vectors with counts from 4 windows, showing just 3 of the potential context word dimensions. The full matrix would have many more dimensions.

Neural Word Embeddings



Neural Sentence Embeddings



Notes on Neural Embeddings

More advanced neural networks are used these days to generate both word and sentence embeddings. The latter using two approaches:

- Static embeddings (fixed vectors, ignore context)
- Contextual embeddings (vector changes depending on surrounding words)

For the inquisitive reader, see this link.

A lecture on the above will be provided in week 9, as part of an overview to artificial neural nets for NLP.

Tools for Neural Embeddings

Static Embeddings:
Glove

Contexual Embeddings:
SentenceTransformer

Multiple choices are available for each of them. We will get to appreciate some of then in the workshop.

Today

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Word Similarity

The similarity between words v and w can be done using the cosine metric, which requires vectors with the same dimensionality.

The cosine metric is based on the dot product:

$$\text{dot product}(\mathbf{v},\mathbf{w}) = \mathbf{v}\cdot\mathbf{w} = \sum_{i=1}^{N} v_i w_i = v_1 w_1 + v_2 w_2 + \ldots + v_N w_N$$

This is fine for normalised vectors (numbers in the range [-1,…,1]), bot not for unnormalised ones.

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Word Similarity

For unnormalised vectors, we use the cosine metric is based on the normalised dot product:

$$\text{cosine}(\mathbf{v},\mathbf{w}) = \frac{\mathbf{v}\cdot\mathbf{w}}{|\mathbf{v}||\mathbf{w}|} = \frac{\sum_{i=1}^{N} v_i w_i}{\sqrt{\sum_{i=1}^{N} v_i^2}\sqrt{\sum_{i=1}^{N} w_i^2}}$$

Outputs of the metric:

- 1: maximal similarity
- 0: no similarity
- $0 < cosine(\mathbf{v},\mathbf{w}) < 1$: partial similarity

Word Similarity: Example

Let's consider the following word counts:

	pie	data	computer
cherry	442	8	2
digital	5	1683	1670
information	5	3982	3325

Examples:

$$\cos(\text{cherry}, \text{information}) = \frac{442*5+8*3982+2*3325}{\sqrt{442^2+8^2+2^2}\sqrt{5^2+3982^2+3325^2}} = .018$$

$$\cos(\text{digital}, \text{information}) = \frac{5*5+1683*3982+1670*3325}{\sqrt{5^2+1683^2+1670^2}\sqrt{5^2+3982^2+3325^2}} = .996$$

Applications of Text Similarity

- Information/document retrieval, to find documents by meaning.
- Question answering, matching a question to the most relevant answer(s).
- Product recommendation, to suggest similar prods.
- Job/applicant search, to match candidates to jobs or vice versa, jobs to applicants.
- Clustering of product/service reviews.
- Etc.

Today

Vector semantics

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Embeddings for Logistic Regression

Given average word embeddings or sentence embeddings, we can use those vectors as features.

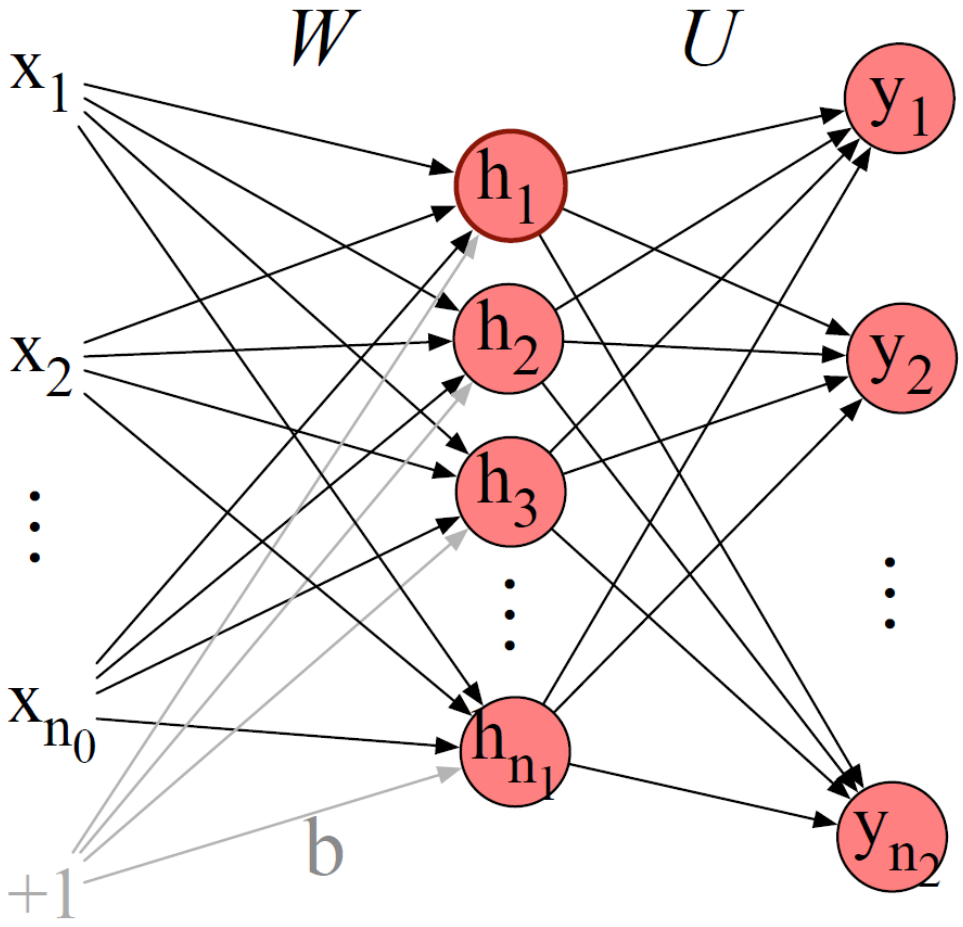
$$
\begin{aligned}
P(+|x) = P(y=1|x) &= \sigma(\mathbf{w}\cdot\mathbf{x}+b) \\
&= \sigma([2.5,-5.0,-1.2,0.5,2.0,0.7]\cdot[3,2,1,3,0,4.19]+0.1) \\
&= \sigma(.833) \\
&= 0.70 \\
P(-|x) = P(y=0|x) &= 1-\sigma(\mathbf{w}\cdot\mathbf{x}+b) \\
&= 0.30
\end{aligned}
$$

$\sigma(x) = \frac{1}{1+e^{-x}}$

dot product

Recall that the weights **w** are learnt by the LR model.

Embeddings for Multi-Layer Perceptrons



x_i represent sentence embeddings, weights **W** and **U** (among others if deeper net) are learnt by the MLP model. The MLP learns a mapping from embeddings to outputs.

Multi-Layer Perceptrons (cont.)

- MLPs are trained using the backpropagation algorithm—not covered in this module.

- The outputs can use the Softmax function to output a probability distribution: $$\text{softmax}(z_i) = \frac{\exp(z_i)}{\sum_{j=1}^{K} \exp(z_j)}$$

- Whilst n-gram classifiers are fully-probabilistic models, logistic regression and MLPs are gradient-based models.

- MLPs are scalable to a large number of inputs & outputs.

Consider scaling your data when using MLPs.

Today

- **Vector semantics**: words as vectors (count-based) to represent word meaning in NLP.
- **Word/Sentence Embeddings**: static, contextual (dense vectors)
- **Text similarity**: high application in NLP tasks.
- **Embedding-based classification**: LR, MLPs, among others.

What's Next?

- Readings:
  - Jurafsky and Martin. Speech and Language Processing. Chapter 5, 2026. [recommended]
  - Almeida & Xexeo. Word Embeddings: A Survey. CoRR, 2019.

- Workshop **tomorrow**: Exercises on text similarity and classification with word/sentence embeddings.

- Next lecture: Large Language Models.